

SDRA: A Spatiotemporally Decoupled RL Agent Enhanced by a Vision-Language Model for Heavy-Duty Vehicle Eco-Driving

Jiwei Yang^{1*}, Zhengxian Chen^{1,2*†}, Pengyi Xiao¹, Weihua Pan¹, Chenggang Zhang¹,
Lei Yang³, Wenfeng Guo⁴, Chaosheng Huang¹, Jun¹

¹School of Vehicle and Mobility, Tsinghua University, Beijing, China

²State Key Laboratory of Intelligent Green Vehicle and Mobility, Tsinghua University, Beijing, China

³School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore

⁴College of Automotive and Energy Engineering, Tongji University, Shanghai, 201804, China

Corresponding author: chenzhengxian@tsinghua.vin

Abstract

Heavy-duty commercial vehicles are characterized by large mass, strong inertia, high fuel consumption, and complex interactions in multi-lane traffic. To improve eco-driving performance and decision stability in complex highway environments, this paper proposes a vision-language-model-enhanced spatio-temporal decoupled reinforcement learning agent. The agent retains a dual-layer architecture consisting of a macro-spatial intent planner and a micro-temporal eco-controller. The vision-language model integrates road images with structured traffic states to identify car-following situations, vehicle cut-ins, traffic congestion, lane opportunities, and potential conflicts, providing scene descriptions, risk assessments, and task-level guidance for lane decision-making and cruise-coast mode selection. Based on this information, the dual-layer policies generate driving decisions, while the safety shield, vehicle dynamics model, and trajectory generator ensure safe and physically feasible vehicle control. Experiments on real-world heavy-duty vehicle data show that the learned policies achieve lower and more concentrated fuel costs than human driving, while the complete spatial-temporal architecture exhibits fewer high-cost outliers. The optimized trajectory reduces fuel consumption by 6.7%. The proposed method therefore improves complex-scene understanding and decision interpretability while maintaining real-time control, operational safety, and energy-saving capability.

Introduction

Heavy-duty commercial vehicles combine large mass, strong inertia, and high fuel demand with complex interactions in multi-lane traffic. Eco-driving can reduce unnecessary acceleration, braking, and power demand, but a practical autonomous system must also understand evolving traffic scenes and select stable maneuvers in real time (International Energy Agency 2017; Huang et al. 2018; Sharma et al. 2021). Model-based eco-driving and look-ahead control provide physically interpretable solutions, yet their performance depends on explicit models and predefined traffic representations (Sciarretta, De Nunzio, and Ojeda 2015; Hellström et al. 2009; Turri et al. 2018).

Deep reinforcement learning (RL) provides a complementary data-driven approach for learning driving policies from

high-dimensional observations (Lillicrap et al. 2016; Fujimoto, Hoof, and Meger 2018; Schulman et al. 2017; Kiran et al. 2022). However, a single policy must otherwise couple lane-level spatial decisions with pedal-level temporal control. For heavy-duty vehicles, these decisions operate at different scales: lane selection and yielding require a global understanding of traffic topology, whereas cruise and coast decisions require frequent longitudinal control. Hierarchical and temporally abstract RL therefore provide a natural basis for separating the two processes (Barto and Mahadevan 2003; Sutton, Precup, and Singh 1999).

A remaining limitation is semantic scene understanding. Structured states such as relative distance and velocity are efficient for control but do not fully express the meaning of a traffic scene, including a developing cut-in, a congested lane, an emerging lane opportunity, or a potential conflict. Vision-language models (VLMs) connect visual observations with language-level concepts and can produce descriptions and task guidance from multimodal inputs (Radford et al. 2021; Alayrac et al. 2022; Li et al. 2023; Liu et al. 2023). Recent autonomous-driving research further demonstrates that language-enhanced models can support interpretable scene reasoning, question answering, and closed-loop driving decisions (Sima et al. 2024; Shao et al. 2024; Xu et al. 2024; Choudhary et al. 2024).

This paper proposes SDRA, a spatiotemporally decoupled RL agent enhanced by a vision-language model for heavy-duty vehicle eco-driving. The VLM jointly processes road images and structured traffic states to judge whether the current traffic moment requires and permits activation of the eco-driving policy. When eco-driving is activated, two co-operating agents generate the detailed maneuver: a Spatial Agent (SA) selects macro-spatial lane intent and a Temporal Agent (TA) selects micro-temporal cruise-coast control. When activation is deferred, the vehicle remains under the conventional safety-preserving driving mode. A safety shield, a vehicle-dynamics model, and a trajectory generator remain active in either case. Thus, the VLM is an upstream semantic trigger, not a direct low-level controller. The system uses a data-anchored training and physics-guided evaluation pipeline grounded in synchronized heavy-duty vehicle mea-

*These authors contributed equally.

†Corresponding author.

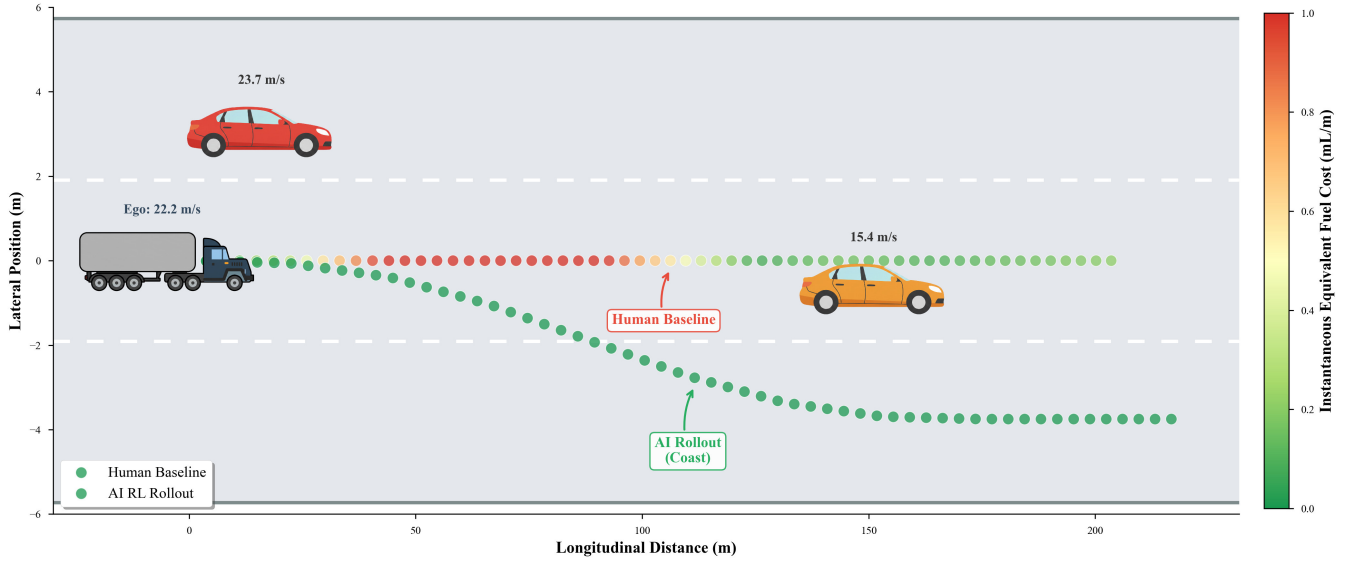


Figure 1: The spatial trajectory and instantaneous equivalent fuel cost of the human-driving baseline and the SDRA rollout.

surements.

The contributions are summarized as follows:

- We establish an industrial-grade, real-world data platform on a 49-ton heavy-duty semi-trailer and construct multimodal observations from synchronized road images, radar, localization, CAN-bus, and engine signals. A VLM interface interprets the scene and determines whether the current moment should activate the eco-driving policy without discarding the measured structured states.
- We develop a spatiotemporally decoupled two-agent RL architecture in which the Spatial Agent handles lane-level intent and the Temporal Agent handles cruise-coast control. Training uses a data-anchored open-loop paradigm via Teacher Forcing, preserving the stochastic evolution and sensor noise of recorded highway traffic instead of replacing them with recursively generated synthetic traffic.
- We formulate a physics-guided spatial evaluation system based on an empirical engine-efficiency map and a kinetic-debt penalty. A three-way comparison—full SA+TA (AI L1+L2), TA-only (AI L2), and the human baseline—quantifies their energy distributions and complementary spatial-temporal roles, while the matched-distance analysis reports the representative 6.7% fuel reduction.

Related Work

Heavy-Duty Vehicle Eco-Driving

Eco-driving has been formulated through optimal control, dynamic speed advisory, and traffic-aware trajectory planning (Barth and Boriboonsomsin 2009; Sciarretta, De Nunzio, and Ojeda 2015; Huang et al. 2018). Heavy-duty-vehicle research emphasizes look-ahead speed control, adaptive cruise control, and platooning because large vehicle

mass makes avoidable speed fluctuations especially costly (Hellström et al. 2009; Turri et al. 2014; Turri, Besselink, and Johansson 2017; Turri et al. 2018; Sharma et al. 2021). Connected-vehicle studies also use predicted behavior of preceding traffic to improve energy-aware control (Ozkan and Ma 2020). These methods establish the physical basis for heavy-truck eco-driving but generally assume a predefined scene representation rather than open-ended semantic interpretation of visual traffic context.

Reinforcement Learning and Spatiotemporal Decoupling

Value-based and actor-critic methods provide the main algorithmic foundations for deep RL control (Sutton and Barto 1998; Watkins and Dayan 1992; Mnih et al. 2015; van Hasselt, Guez, and Silver 2016; Silver et al. 2014; Haarnoja et al. 2018). Continuous-control and policy-optimization algorithms have been widely adopted in autonomous-driving research (Lillicrap et al. 2016; Fujimoto, Hoof, and Meger 2018; Schulman et al. 2015, 2017). Energy-aware driving studies demonstrate that RL can learn eco-driving behavior and hierarchical energy-management strategies (Qi et al. 2019; Wu et al. 2023). Hierarchical RL and temporal abstraction further motivate separating macro-level intent from local execution (Barto and Mahadevan 2003; Sutton, Precup, and Singh 1999). Teacher Forcing and scheduled sampling address how sequential models are exposed to recorded versus self-generated histories (Williams and Zipser 1989; Bengio et al. 2015); SDRA uses the recorded-data side of this principle to anchor the exogenous traffic evolution during open-loop policy training. The framework adopts this separation specifically for lane-level spatial reasoning and cruise-coast temporal control.

Vision-Language Models for Driving

Large-scale vision-language pre-training enables models to align images with transferable language concepts (Radford et al. 2021; Alayrac et al. 2022; Li et al. 2023; Liu et al. 2023). For autonomous driving, DriveLM represents driving reasoning through graph-based visual question answering, Talk2BEV introduces language-enhanced bird’s-eye-view representations, and LMDrive and DriveGPT4 study language-guided or interpretable closed-loop driving (Sima et al. 2024; Choudhary et al. 2024; Shao et al. 2024; Xu et al. 2024). Unlike approaches that allow a large model to directly generate low-level control, SDRA uses the VLM as an eco-driving activation gate. It judges whether the current scene calls for energy-optimizing intervention; after activation, real-time lane actions remain the responsibility of the Spatial Agent and cruise-coast actions remain the responsibility of the Temporal Agent.

Safe and Physically Feasible Execution

Autonomous-driving systems commonly separate behavior planning from motion generation and vehicle control (Schwartz, Alonso-Mora, and Rus 2018; Dolgov et al. 2010; Werling et al. 2010; Polack et al. 2017). Safe RL and shielding methods constrain learned actions before execution (Achiam et al. 2017; Alshiekh et al. 2018). SDRA follows this modular principle: the VLM decides whether to invoke eco-driving, the Spatial Agent and Temporal Agent select decisions after invocation, and the safety shield and dynamics-based trajectory generator determine whether and how those decisions are executed.

Multimodal Inputs and Traffic Scene Definition

Real-World Heavy-Duty Vehicle Data Platform

The empirical data are collected from a heavy-duty semi-trailer operating in highway traffic. The vehicle has a fully loaded mass of 49,000 kg and a frontal area of 10 m². The physical model uses an aerodynamic drag coefficient of 0.52, a tire rolling-resistance coefficient of 0.006, an air density of 1.225 kg/m³, and gravitational acceleration of 9.81 m/s². These parameters preserve the inertia and resistance characteristics that distinguish heavy-duty eco-driving from passenger-vehicle control.

The sensing platform contains a front-bumper dual-mode millimeter-wave radar, a windshield-mounted road camera, GPS, IMU/ESC sensors, and direct access to the chassis CAN bus. The radar provides a 120° near-field view up to 70 m and an 18° far-field view up to 250 m, while the camera supplies lane and road context up to approximately 200 m. The localization sensors provide position, heading, and ego-motion. The CAN bus records vehicle speed, throttle and brake openings, engine speed, and engine torque. All streams are synchronized so that each road image is paired with structured traffic and vehicle states.

Road Images and Structured Traffic States

At decision time t , SDRA receives a road image I_t and a structured traffic state x_t . The image supplies visual context

for lane geometry, vehicle appearance, occlusion, and scene layout. The structured state contains the ego-vehicle speed and acceleration, lane index, relative distance and velocity of surrounding vehicles, available gaps, and vehicle-dynamics variables required by the Temporal Agent. Combining the two modalities allows the system to retain precise numerical information while introducing semantic information that is difficult to encode through fixed state variables alone.

The multimodal observation is written as $o_t = (I_t, x_t)$. The VLM transforms o_t into a semantic representation z_t containing a scene description, a risk assessment, and task-level guidance. Its primary control output is an eco-driving activation variable $q_t \in \{0, 1\}$, where $q_t = 1$ invokes the SA-TA eco-driving policy and $q_t = 0$ defers activation. The VLM output is therefore a mode-selection decision rather than an actuator command.

Traffic Scene Categories and Semantic Tasks

The semantic reasoning task focuses on the traffic situations specified in the abstract: car following, vehicle cut-ins, traffic congestion, lane opportunities, and potential conflicts. For each observation, the VLM identifies the dominant scene category, summarizes the relevant interacting vehicles, assigns a qualitative risk level, and answers the operational question: should the eco-driving policy be activated at this moment? Activation is favored when anticipatory spatial or temporal action can avoid a likely reactive deceleration–acceleration cycle, while activation is deferred when the scene is routine, ambiguous, stale, or incompatible with the safety conditions.

Figure 2 illustrates the vehicle response associated with a representative cut-in. Around the 17-second mark, the recorded brake-pedal spike and speed drop reveal a high-energy reactive response. This event motivates the activation task: the VLM should recognize when an emerging interaction creates an opportunity for anticipatory eco-driving before the driver enters a braking and recovery cycle.

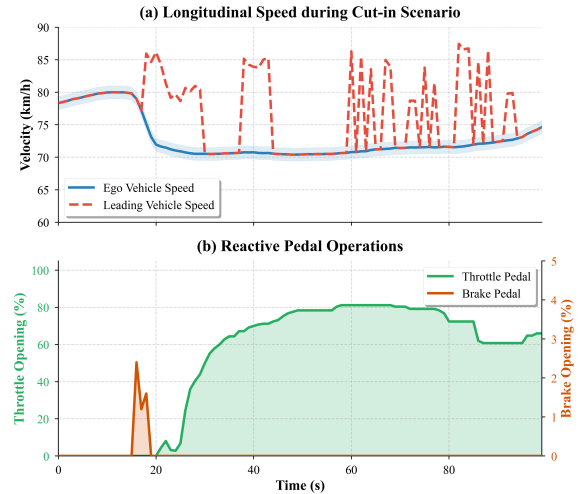


Figure 2: The longitudinal speed and pedal opening during a representative cut-in scenario.

Spatio-Temporal Decoupled Eco-Driving Framework

System Architecture and Problem Formulation

SDRA formulates heavy-duty vehicle eco-driving as a VLM-gated two-agent Markov decision process. At time t , the system observes a road image I_t and a structured traffic state x_t . The VLM first produces semantic reasoning z_t and an activation decision q_t . Only when $q_t = 1$ does the Spatial Agent with policy π_{SA} select the macro-spatial intent and the Temporal Agent with policy π_{TA} select the micro-temporal cruise-coast mode. The decision process is

$$\begin{aligned} z_t &= f_{VLM}(I_t, x_t), \\ q_t &= G(z_t) \in \{0, 1\}, \\ (a_t^{spatial}, a_t^{temporal}) &\sim (\pi_{SA}, \pi_{TA}), \quad q_t = 1, \\ a_t &= a_t^{default}, \quad q_t = 0. \end{aligned}$$

Here, $a_t^{default}$ denotes the conventional safety-preserving driving mode. The dual-agent control and physical-execution backbone is shown in Fig. 3. Semantic gating, eco-driving decisions, and physical execution are separated: the VLM decides whether the energy-optimizing policy is needed, the SA and TA determine how to drive after activation, and the safety and dynamics modules determine what can be executed.

State Space Representation

The structured state contains ego-vehicle dynamics and the surrounding traffic topology. It includes longitudinal velocity, acceleration, lane index, engine-related variables, and the relative distance and velocity of relevant surrounding vehicles. The visual state contains the forward road image and lane-level appearance cues. The VLM output is represented as $z_t = [e_t^{scene}, r_t^{risk}, g_t^{task}, c_t]$, where e_t^{scene} summarizes the scene, r_t^{risk} is the estimated risk level, g_t^{task} explains why eco-driving is or is not appropriate, and c_t is confidence. A deterministic parser maps this output to q_t . Language is therefore converted into a bounded machine-readable activation signal before control proceeds.

Vision-Language Eco-Driving Activation

The VLM prompt is organized around four questions: what is happening in the traffic scene, which objects are relevant to the heavy-duty vehicle, whether reactive braking or inefficient recovery is likely over the next decision horizon, and whether the SA-TA eco-driving policy should be activated now. The response is parsed into a fixed schema containing the scene label, relevant interacting objects, risk level, activation recommendation, confidence, and a concise rationale.

To maintain real-time control, the VLM operates at a lower semantic update frequency than the RL control loop. Its latest valid output is cached and reused only within a bounded freshness interval. Low-confidence, stale, unavailable, or schema-invalid outputs set $q_t = 0$, so the VLM cannot activate eco-driving on uncertain evidence. The safety shield remains authoritative even when $q_t = 1$.

Data-Anchored Open-Loop Training via Teacher Forcing

The SA and TA use a data-anchored open-loop training paradigm. At each training step, the surrounding-traffic observations and heterogeneous sensor context are anchored to the synchronized next sample in the recorded highway sequence rather than recursively replaced by a synthetic traffic rollout. This use of recorded histories follows the Teacher Forcing principle of conditioning sequential learning on observed data (Williams and Zipser 1989; Bengio et al. 2015). Candidate SA and TA actions are evaluated through the reward, safety, vehicle-dynamics, and trajectory-generation modules, while the exogenous traffic evolution preserves the naturalistic stochasticity and sensor noise of the industrial data. The VLM gate is evaluated on the same recorded image-state pair.

Hierarchical Interface and Execution Delegation

When $q_t = 1$, the Spatial Agent (SA) chooses among keep-lane, left-lane, and right-lane intents from the structured traffic state. The VLM does not choose among these actions; its task-level rationale is retained as advisory context for interpreting the decision, while its control authority ends at deciding whether the eco-driving policy should be invoked.

After the spatial intent is selected, a mask Φ extracts the target-lane traffic context for the Temporal Agent (TA):

$$s_t^{lower} = \Phi(x_t, a_t^{spatial}),$$

so the TA receives the precise measurements for the lane selected by the SA. This hard-attention interface prevents an unverified language output from directly changing the low-level control state.

The agents are cooperative rather than independent. The SA sends an inter-agent message

$$h_t^{SA} = [l_t^{target}, c_t^{lane}, H_t^{intent}],$$

containing the target lane, lane-intent confidence, and intended decision horizon. The TA conditions its cruise-coast decision on this message and returns a feasibility indicator and predicted longitudinal response. If the TA or safety shield determines that the requested spatial intent cannot be supported by a safe longitudinal profile, the SA retains the current lane or selects a conservative alternative at the next spatial decision step.

The Temporal Agent outputs two discrete temporal modes, cruise and coast. These modes are passed to a deterministic longitudinal model that converts the decision into a physically feasible acceleration command. In coast mode, the vehicle decelerates under modeled mechanical resistance. In cruise mode, it maintains the desired speed in free flow and uses a parameterized Intelligent Driver Model (IDM) when following a preceding vehicle:

$$a_{exec} = \begin{cases} -\frac{F_{res}(v_t)}{m}, & m_t = m_{coast} \\ 0, & m_t = m_{cruise} \wedge \text{Free-flow} \\ a_{IDM}, & m_t = m_{cruise} \wedge \text{Car-following} \end{cases}$$

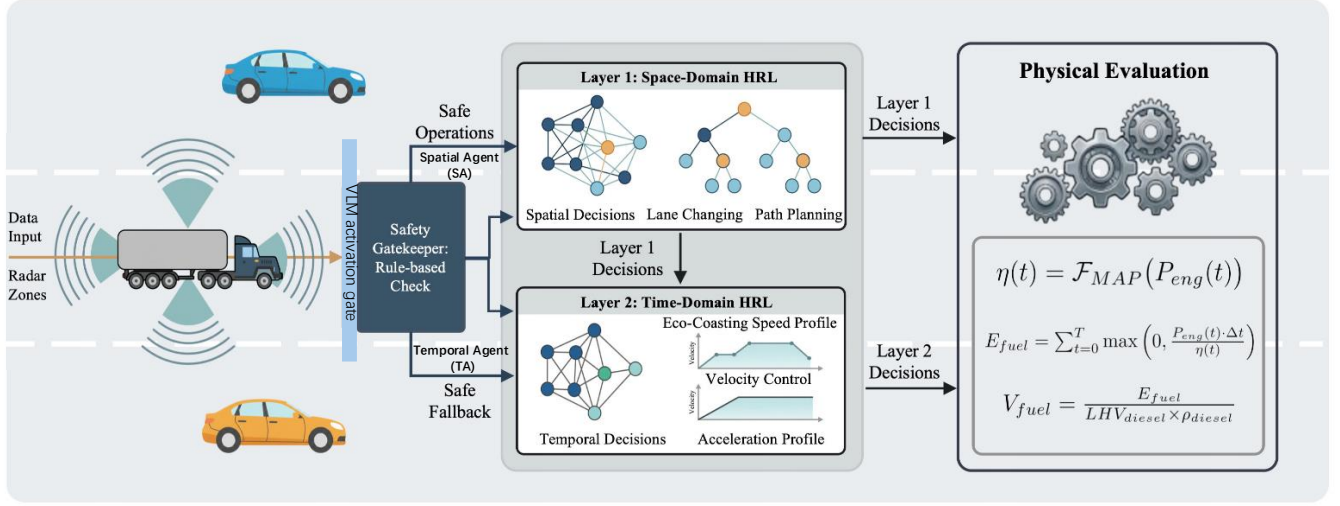


Figure 3: The two-agent RL and physics-based execution backbone of SDRA. The VLM serves as an upstream eco-driving activation gate; after activation, the Spatial Agent and Temporal Agent generate the spatial and temporal decisions.

where the car-following acceleration a_{IDM} is given by:

$$a_{IDM} = a_{max} \left[1 - \left(\frac{v_t}{v_0} \right)^\delta - \left(\frac{s^*}{\Delta x_{target}} \right)^2 \right]$$

Here, v_0 is the target velocity, a_{max} is the maximum acceleration, and s^* is the desired dynamic gap. The delegated execution model keeps the learned policy focused on strategic energy behavior while providing a deterministic mechanism for generating longitudinal motion (Treiber, Hennecke, and Helbing 2000; Kesting, Treiber, and Helbing 2010).

Hierarchical Reward Formulation

Because SDRA decouples spatial routing and temporal speed regulation, separate rewards train the Spatial Agent and Temporal Agent. The VLM activation gate does not directly supervise either agent's action.

Macro-Spatial Routing Reward (Layer 1) The Spatial Agent aims to allocate right-of-way in multi-lane traffic while discouraging unnecessary lane changes. The spatial reward $R_{spatial}$ balances traffic efficiency, spatial opportunity, and maneuver stability.

To induce long-horizon anticipatory behaviors, the framework continuously monitors the ego-lane gap Δx_{ego} . A continuous congestion anxiety penalty is applied when this gap falls below a proactive threshold $d_{anx} = 150$ m:

$$R_{anx} = -\alpha \max(0, d_{anx} - \Delta x_{ego})$$

where α is a scaling coefficient.

When the agent decides to maintain the current lane ($a_{spatial} = d_{keep}$), it receives a baseline stability bonus b_{keep} to anchor the vehicle in empty lanes. However, to prevent passive driving, a missed-opportunity penalty is imposed if adjacent lanes offer significantly more driving space:

$$R_{spatial}(d_{keep}) = R_{anx} + b_{keep} - \gamma_{miss}(\Delta x_{max} - \Delta x_{ego})$$

where Δx_{max} is the maximum available gap across all lanes, and γ_{miss} penalizes the failure to exploit optimal right-of-way.

Conversely, executing a lane change ($a_{spatial} \in \{d_{left}, d_{right}\}$) requires overcoming a fixed physical maneuver cost C_{lc} to prevent undesirable oscillation. The reward evaluates the actual spatial gain against this fixed cost, subject to a stringent safety boundary penalty p_{danger} :

$$R_{spatial}(d_{change}) = R_{anx} - C_{lc} + \gamma_{gain}(\Delta x_{target} - \Delta x_{ego}) - p_{danger} \max(0, d_{safe} - \Delta x_{target})$$

where Δx_{target} is the gap in the intended lane, and $d_{safe} = 100$ m. This formulation ensures that a lane change is rewarded only when its spatial gain outweighs maneuver cost and safety risk.

Micro-Temporal Coasting Reward (Layer 2) Once the SA selects a spatial intent, the Temporal Agent outputs binary mode commands (m_{cruise}, m_{coast}). A context-aware asymmetric reward partitions the environment into a safe state $\mathcal{S}_{safe}$ and a threat state $\mathcal{S}_{danger}$ using structured time-gap and relative-speed conditions.

The asymmetric reward is defined by a state-dependent piecewise mechanism:

$$R_{temporal}(s_t, a_t) = \begin{cases} r_{defend}, & s_t \in \mathcal{S}_{danger}, m_t = m_{coast} \\ -p_{hazard}, & s_t \in \mathcal{S}_{danger}, m_t = m_{cruise} \\ r_{cruise}, & s_t \notin \mathcal{S}_{danger}, m_t = m_{cruise} \\ -p_{ineff}, & s_t \notin \mathcal{S}_{danger}, m_t = m_{coast} \end{cases}$$

By rewarding defensive coasting in hazardous states ($r_{defend} = 15$) and penalizing cruising in those states ($-p_{hazard} = -50$), the reward encourages safe early coasting.

Safety Shield and Dynamics-Based Execution

Before execution, the safety shield checks the proposed lane intent and cruise-coast mode against lane availability, relative gaps, time-to-collision conditions, and vehicle-dynamics limits. An action that violates the predefined constraints is rejected or replaced by a conservative fallback. This shield follows the separation between learned decision making and constrained execution used in safe RL (Achiam et al. 2017; Alshiekh et al. 2018).

Accepted spatial intents are converted into continuous trajectories with a quintic polynomial generator that matches position, velocity, and acceleration boundary conditions (Werling et al. 2010). Longitudinal commands are generated by the resistance and car-following models described above. The combined vehicle model and trajectory generator limit acceleration, deceleration, and jerk so that the output remains physically feasible for a heavy-duty vehicle.

Energy use is evaluated from the generated speed, acceleration, and engine operating trajectory. The required tractive power includes inertial, aerodynamic, and rolling-resistance components, and fuel use is obtained through the engine-efficiency map. The VLM only decides when to invoke the eco-driving policy; it neither estimates fuel consumption nor replaces the physical energy model.

To compare trajectories over the same traveled distance, the physics-guided spatial fuel metric is used. Instantaneous required power $P_{req}(t)$ is mapped through the empirical engine-efficiency function $\eta = f_{map}(P_{req})$ to obtain operational fuel consumption $F_{ops,t}$. A kinetic-debt term $F_{debt,t}$ represents the fuel-equivalent cost of dissipated kinetic energy or a terminal velocity deficit. The equivalent spatial fuel cost is

$$C_{spatial} = \frac{\sum_{t=0}^T (F_{ops,t} + F_{debt,t}) \times 1000}{\sum_{t=0}^T v_t \Delta t},$$

measured in mL/m. This metric provides a consistent physical comparison for the VLM-gated and structured-state agents under the same energy model.

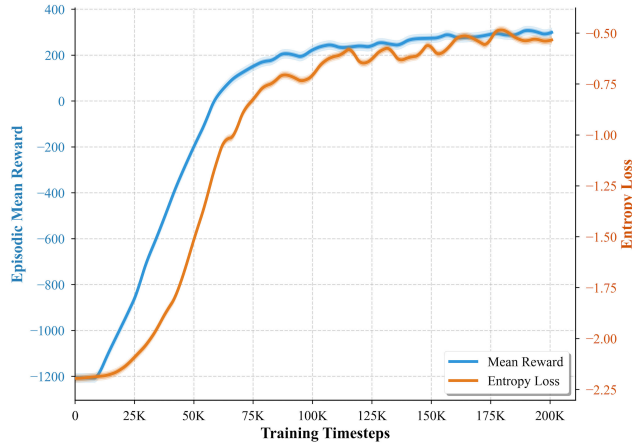


Figure 4: The episodic mean reward and policy entropy loss during the training process.

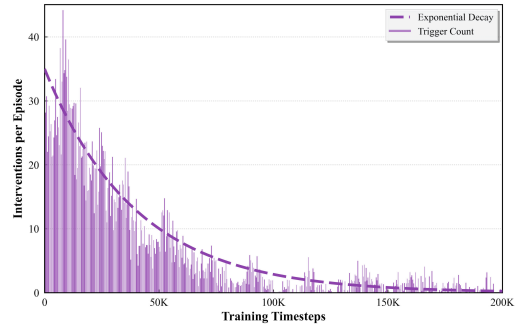


Figure 5: The safety shield intervention rate per episode during the training process.

Experiments and Results

Experimental Setup

The framework is evaluated using the 49-ton heavy-duty truck highway dataset. Both the Spatial Agent and Temporal Agent are optimized using PPO with an actor-critic architecture (Schulman et al. 2017). Training directly on real-world heterogeneous sensor streams exposes the policies to naturalistic traffic stochasticity and sensor noise.

Training Dynamics and Policy Convergence

Training dynamics are evaluated using the episodic reward and policy entropy shown in Fig. 4. Within the initial 75,000 steps, the mean episodic reward rises from approximately -1200 to the positive range, while the entropy loss stabilizes. Together, these curves indicate a transition from broad exploration toward a more stable policy.

Figure 5 reports the number of guardrail interventions during training. In the preliminary phase, the guardrail triggers more than 40 times per episode as the policy explores hazardous actions. The intervention frequency decreases and approaches zero by approximately 150,000 steps, indicating that the learned policy increasingly produces actions that satisfy the predefined execution constraints.

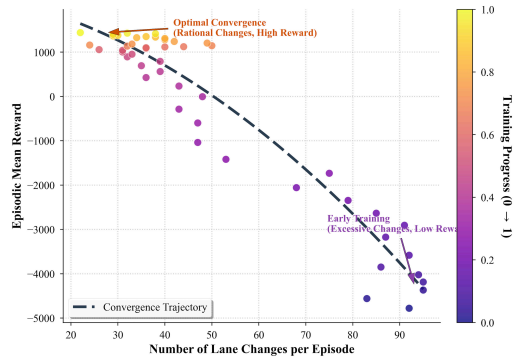


Figure 6: The episodic mean reward and number of lane changes across different training stages.

Figure 6 shows the evolution of the lane-change count across training stages. The early policy exhibits irrational

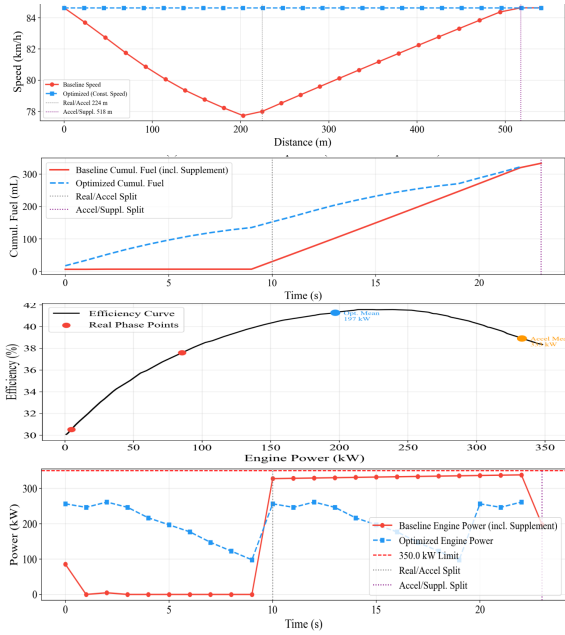


Figure 7: The cumulative fuel consumption, vehicle speed, engine efficiency, and engine power across the baseline and optimized driving profiles. From top to bottom: (a) cumulative fuel comparison (baseline vs. optimized); (b) distance-axis speed comparison (baseline vs. optimized, with the same total distance); (c) engine efficiency curve and operating points; and (d) full-timeline engine power comparison (baseline vs. optimized).

lateral exploration, reaching as many as approximately 100 lane changes per episode and incurring substantial kinematic penalties. After convergence, the count stabilizes at approximately 20–40 lane changes per episode, consistent with the Spatial Agent learning to exploit useful lane opportunities while avoiding unnecessary oscillation.

Fuel Economy Benchmark and Behavioral Analysis

To objectively evaluate the thermodynamic advantages of the decoupled framework, the generated trajectories are compared against real human-driving segments using the physics-guided spatial metric $C_{spatial}$ (mL/m). The evaluation incorporates the engine-efficiency map and a kinetic-debt penalty for terminal velocity deficits.

As shown in Fig. 8, the human baseline exhibits highly dispersed and elevated equivalent fuel costs under different initial gap distances. Both AI configurations (AI L1+L2 and AI L2) maintain a lower, concentrated level of approximately 0.3 mL/m. The inset box plot reports a highly significant difference (****) between the human baseline and both AI models. Although the average spatial fuel cost difference between AI L1+L2 and pure AI L2 is not significant (ns), AI L1+L2 has a more concentrated distribution with visibly fewer high-cost outliers. The pure L2 controller preserves longitudinal momentum but remains confined to its lane, whereas L1+L2 can bypass downstream bottlenecks through

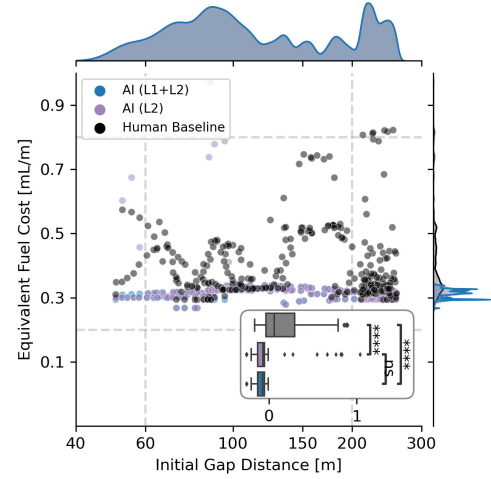


Figure 8: The equivalent fuel cost and initial gap distance across human and AI agents.

strategic lane changes, improving global mobility without compromising the thermodynamic baseline established by L2.

In the representative car-following case, the approximately 41-ton semi-trailer starts at 84.64 km/h (23.51 m/s), while human braking lowers speed to 77.74 km/h (21.59 m/s) over 10 s. Both trajectories cover 540.75 m: a 224.4 m recorded deceleration phase, a 293.2 m re-acceleration phase over 13 s, and a 23.2 m constant-speed supplement. The human profile consumes approximately 6.83 mL during the initial 9 s but reaches 333.7 mL at 24 s after high-power recovery; the optimized profile reaches 311.4 mL at 23 s. Its average power and efficiency are 197 kW and 40.76%, compared with 333 kW and 38.92% during human re-acceleration. This 1.8 percentage-point efficiency difference contributes to a 6.7% fuel reduction in the matched-distance comparison.

Conclusion

This paper presented SDRA, a VLM-enhanced spatiotemporally decoupled reinforcement learning framework for heavy-duty vehicle eco-driving. The VLM serves as an upstream semantic gate, while the Spatial Agent selects lane-level intent and the Temporal Agent performs cruise-coast control. This separation enables scene understanding, strategic planning, and real-time execution to operate at appropriate spatial and temporal scales. The safety shield, vehicle-dynamics model, trajectory generator, and data-anchored Teacher Forcing strategy keep the learned decisions safe, physically feasible, and grounded in naturalistic traffic. Experiments using real-world heavy-duty vehicle data show a lower and more concentrated fuel-cost distribution than human driving, fewer high-cost outliers with the complete architecture, and a representative fuel reduction of 6.7%. These findings demonstrate that multimodal semantic reasoning can effectively trigger physics-guided eco-driving without directly controlling the vehicle. Future work will evaluate VLM activation accuracy,

latency, and uncertainty across more diverse traffic, weather, road, and vehicle-loading conditions.

References

- Achiam, J.; Held, A.; Tamar, A.; and Abbeel, P. 2017. Constrained Policy Optimization. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, 22–31.
- Alayrac, J.-B.; Donahue, J.; Luc, P.; Miech, A.; Barr, I.; Hasson, Y.; Lenc, K.; Mensch, A.; Millican, K.; Reynolds, M.; et al. 2022. Flamingo: A Visual Language Model for Few-Shot Learning. In *Advances in Neural Information Processing Systems*, volume 35, 23716–23736.
- Alshiekh, M.; Bloem, R.; Ehlers, R.; Könighofer, B.; Niekum, S.; and Topcu, U. 2018. Safe Reinforcement Learning via Shielding. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1).
- Badue, C.; Guidolini, R.; Carneiro, R. V.; A., A. M.; et al. 2021. Self-Driving Cars: A Survey. *Expert Systems with Applications*, 165: 113816.
- Bansal, M.; Krizhevsky, A.; and Ogale, A. 2019. Chauffeur-Net: Learning to Drive by Imitating the Best and Synthesizing the Worst. *arXiv preprint arXiv:1812.03079*.
- Barth, M.; and Boriboonsomsin, K. 2009. Energy and Emissions Impacts of a Freeway-Based Dynamic Eco-Driving System. *Transportation Research Part D: Transport and Environment*, 14(6): 400–410.
- Barto, A. G.; and Mahadevan, S. 2003. Recent Advances in Hierarchical Reinforcement Learning. *Discrete Event Dynamic Systems*, 13(1–2): 41–77.
- Bellemare, M. G.; Dabney, W.; and Munos, R. 2017. A Distributional Perspective on Reinforcement Learning. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, 449–458.
- Bengio, S.; Vinyals, O.; Jaitly, N.; and Shazeer, N. 2015. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. In *Proceedings of the 28th International Conference on Neural Information Processing Systems*, 1171–1179.
- Bojarski, M.; Del Testa, D.; Dworakowski, D.; Firner, B.; Flepp, B.; Goyal, P.; Jackel, L. D.; Monfort, M.; Muller, U.; Zhang, J.; et al. 2016. End to End Learning for Self-Driving Cars. *arXiv preprint arXiv:1604.07316*.
- Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Finn, C.; Gopalakrishnan, K.; Hausman, K.; Herzog, A.; Hsu, D.; Ibarz, J.; et al. 2023. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. *arXiv preprint arXiv:2307.15818*.
- Chen, C.; Seff, A.; Kornhauser, A.; and Xiao, J. 2015. Deep-Driving: Learning Affordance for Direct Perception in Autonomous Driving. In *2015 IEEE International Conference on Computer Vision (ICCV)*, 2722–2730.
- Chen, L.; Lu, K.; Rajeswaran, A.; Lee, K.; Grover, A.; Laskin, M.; Abbeel, P.; Srinivas, A.; and Mordatch, I. 2021. Decision Transformer: Reinforcement Learning via Sequence Modeling. In *Advances in Neural Information Processing Systems*, volume 34.
- Chitta, K.; Prakash, A.; Jaeger, A.; Yu, Z.; Renz, K.; and Geiger, A. 2022. TransFuser: Imitation with Transformer-Based Sensor Fusion for Autonomous Driving. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Choudhary, T.; Dewangan, V.; Chandhok, S.; Priyadarshan, S.; Jain, A.; Singh, A. K.; Srivastava, S.; Jatavallabhula, K. M.; and Krishna, K. M. 2024. Talk2BEV: Language-Enhanced Bird’s-Eye View Maps for Autonomous Driving. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 16345–16352.
- Cordts, M.; Omran, M.; Ramos, S.; Rehfeld, T.; Enzweiler, M.; Benenson, R.; Franke, U.; Roth, S.; and Schiele, B. 2016. The Cityscapes Dataset for Semantic Urban Scene Understanding. In *2016 IEEE Conference on Computer Vision and Pattern Recognition*, 3213–3223.
- Dolgov, D.; Thrun, S.; Montemerlo, M.; and Diebel, J. 2010. Path Planning for Autonomous Vehicles in Unknown Semi-Structured Environments. *The International Journal of Robotics Research*, 29(5): 485–501.
- Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; et al. 2021. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*.
- Driess, D.; Xia, F.; Sajjadi, M. S. M.; Lynch, C.; Chowdhery, A.; Ichter, B.; Wahid, A.; Tompson, J.; Vuovic, D.; Walkenbach, M.; et al. 2023. PaLM-E: An Embodied Multimodal Language Model. *arXiv preprint arXiv:2303.03378*.
- Finn, C.; Abbeel, P.; and Levine, S. 2017. Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, 1126–1135.
- Fujimoto, S.; Hoof, H.; and Meger, D. 2018. Addressing Function Approximation Error in Actor-Critic Methods. In *Proceedings of the 35th International Conference on Machine Learning*, 1587–1596.
- Geiger, A.; Lenz, P.; and Urtasun, R. 2012. Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite. In *2012 IEEE Conference on Computer Vision and Pattern Recognition*, 3354–3361.
- Goodfellow, I.; Bengio, Y.; and Courville, A. 2016. *Deep Learning*. MIT Press.
- Grigorescu, S.; Trasnea, B.; Cocias, T.; and Macesanu, G. 2020. A Survey of Deep Learning Techniques for Autonomous Driving. *Journal of Field Robotics*, 37(3): 362–386.
- Haarnoja, T.; Zhou, A.; Abbeel, P.; and Levine, S. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, 1861–1870.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.

- Hellström, E.; Ivarsson, M.; Åslund, J.; and Nielsen, L. 2009. Look-Ahead Control for Heavy Trucks to Minimize Trip Time and Fuel Consumption. *Control Engineering Practice*, 17(2): 245–254.
- Hessel, M.; Modayil, J.; van Hasselt, H.; Schaul, T.; Ostrovski, G.; Dabney, W.; Horgan, D.; Piot, B.; Azar, M. G.; and Silver, D. 2018. Rainbow: Combining Improvements in Deep Reinforcement Learning. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, volume 32.
- Hochreiter, S.; and Schmidhuber, J. 1997. Long Short-Term Memory. *Neural Computation*, 9(8): 1735–1780.
- Hu, Y.; Yang, J.; Chen, L.; Li, K.; Sima, C.; Zhu, X.; et al. 2023. UniAD: Planning-Oriented Autonomous Driving. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17853–17862.
- Huang, Y.; Ng, E. C. Y.; Zhou, J. L.; Surawski, N. C.; Chan, E. F. C.; and Hong, G. 2018. Eco-Driving Technology for Sustainable Road Transport: A Review. *Renewable and Sustainable Energy Reviews*, 93: 596–609.
- International Energy Agency. 2017. *The Future of Trucks: Implications for Energy and the Environment*. Paris: OECD Publishing.
- Janai, J.; Güney, F.; Behl, A.; and Geiger, A. 2020. Computer Vision for Autonomous Vehicles: Problems, Datasets and State of the Art. *Foundations and Trends in Computer Graphics and Vision*, 12(1–3): 1–308.
- Janner, M.; Li, Q.; and Levine, S. 2021. Offline Reinforcement Learning as One Big Sequence Modeling Problem. In *Advances in Neural Information Processing Systems*, volume 34.
- Jiang, B.; Shao, Z.; et al. 2023. VAD: Vectorized Scene Representation for Efficient Autonomous Driving. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*.
- Kendall, A.; Hawke, A.; Janz, D.; Mazur, P.; Reda, D.; Allen, J.; Lam, V.-D.; Bewley, A.; and Shah, A. 2019. Learning to Drive in a Day. In *2019 International Conference on Robotics and Automation (ICRA)*, 8248–8254.
- Kesting, A.; Treiber, M.; and Helbing, D. 2010. Enhanced Intelligent Driver Model to Access the Impact of Driving Strategies on Traffic Capacity. *Philosophical Transactions of the Royal Society A*, 368(1928): 4585–4605.
- Kingma, D. P.; and Ba, J. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*.
- Kiran, B. R.; Sobh, I.; Talpaert, V.; Mannion, P.; Al Sallab, A. A.; Yogamani, S.; and Perez, P. 2022. Deep Reinforcement Learning for Autonomous Driving: A Survey. *IEEE Transactions on Intelligent Transportation Systems*, 23(6): 4909–4926.
- Kober, J.; Bagnell, J. A.; and Peters, J. 2013. Reinforcement Learning in Robotics: A Survey. *The International Journal of Robotics Research*, 32(11): 1238–1274.
- Kumar, A.; Zhou, A.; Tucker, G.; and Levine, S. 2020. Conservative Q-Learning for Offline Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 33.
- Levine, S.; Finn, C.; Darrell, T.; and Abbeel, P. 2016. End-to-End Training of Deep Visuomotor Policies. *Journal of Machine Learning Research*, 17(39): 1–40.
- Levine, S.; Kumar, A.; Tucker, G.; and Fu, J. 2020. Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems. *arXiv preprint arXiv:2005.01643*.
- Li, J.; Li, D.; Savarese, S.; and Hoi, S. 2023. BLIP-2: Bootstrapping Language-Image Pre-Training with Frozen Image Encoders and Large Language Models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, 19730–19742.
- Li, Z.; Wang, W.; Li, H.; Sima, C.; Lu, J.; Qiao, J.; Lu, L.; Chen, J.; Che, L.; Xu, J.; et al. 2022. BEVFormer: Learning Bird’s-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers. In *Computer Vision – ECCV 2022*.
- Lillicrap, T. P.; Hunt, J. J.; Pritzel, A.; Heess, N.; Erez, T.; Tassa, Y.; Silver, D.; and Wierstra, D. 2016. Continuous Control with Deep Reinforcement Learning. *International Conference on Learning Representations*.
- Liu, H.; Li, C.; Wu, Q.; and Lee, Y. J. 2023. Visual Instruction Tuning. In *Advances in Neural Information Processing Systems*, volume 36, 34892–34916.
- Liu, Z.; Tang, H.; Amini, A.; Yang, X.; Mao, H.; Rus, D.; and Han, S. 2022. BEVFusion: A Simple and Robust LiDAR-Camera Fusion Framework. In *Advances in Neural Information Processing Systems*, volume 35.
- Mnih, V.; Kavukcuoglu, K.; Silver, D.; Rusu, A.; Veness, J.; Bellemare, M.; Graves, A.; Riedmiller, M.; Fidjeland, A.; Ostrovski, G.; et al. 2015. Human-level Control through Deep Reinforcement Learning. *Nature*, 518(7540): 529–533.
- Ng, A. Y.; Harada, D.; and Russell, S. 1999. Policy Invariance Under Reward Transformations: Theory and Application to Reward Shaping. In *Proceedings of the Sixteenth International Conference on Machine Learning*, 278–287.
- Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C.; Mishkin, P.; Zhang, C.; Agarwal, S.; Slama, K.; Ray, A.; et al. 2022. Training Language Models to Follow Instructions with Human Feedback. In *Advances in Neural Information Processing Systems*, volume 35.
- Ozkan, M. F.; and Ma, Y. 2020. Eco-Driving of Connected and Automated Vehicle With Preceding Driver Behavior Prediction. *Journal of Dynamic Systems, Measurement, and Control*, 143(1).
- Paden, B.; Cáp, M.; Yong, S.; Yershov, D.; and Frazzoli, E. 2016. A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles. *IEEE Transactions on Intelligent Vehicles*, 1(1): 33–55.
- Polack, P.; Altché, F.; d’Andréa Novel, B.; and de La Fortelle, A. 2017. The Kinematic Bicycle Model: A Consistent Model for Planning Feasible Trajectories for Autonomous Vehicles? In *2017 IEEE Intelligent Vehicles Symposium (IV)*, 812–818.
- Pomerleau, D. A. 1989. ALVINN: An Autonomous Land Vehicle in a Neural Network. In *Advances in Neural Information Processing Systems*, volume 1.

- Qi, X.; Luo, Y.; Wu, G.; Boriboonsomsin, K.; and Barth, M. 2019. Deep Reinforcement Learning Enabled Self-Learning Control for Energy Efficient Driving. *Transportation Research Part C: Emerging Technologies*, 99: 67–81.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models from Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, 8748–8763.
- Schaul, T.; Quan, J.; Antonoglou, I.; and Silver, D. 2016. Prioritized Experience Replay. In *International Conference on Learning Representations*.
- Schulman, J.; Levine, S.; Abbeel, P.; Jordan, M.; and Moritz, P. 2015. Trust Region Policy Optimization. In *Proceedings of the 32nd International Conference on Machine Learning*, 1889–1897.
- Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347*.
- Schwarting, W.; Alonso-Mora, J.; and Rus, D. 2018. Planning and Decision-Making for Autonomous Vehicles. *Annual Review of Control, Robotics, and Autonomous Systems*, 1: 187–210.
- Sciarretta, A.; De Nunzio, G.; and Ojeda, L. L. 2015. Optimal Ecodriving Control: Energy-Efficient Driving of Road Vehicles as an Optimal Control Problem. *IEEE Control Systems*, 35(5): 71–90.
- Shao, H.; Hu, Y.; Wang, L.; Song, G.; Waslander, S. L.; Liu, Y.; and Li, H. 2024. LMDrive: Closed-Loop End-to-End Driving with Large Language Models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15120–15130.
- Sharma, N. K.; Hamednia, A.; Murgovski, N.; Gelso, E. R.; and Sjöberg, J. 2021. Optimal Eco-Driving of a Heavy-Duty Vehicle Behind a Leading Heavy-Duty Vehicle. *IEEE Transactions on Intelligent Transportation Systems*, 22(12): 7792–7803.
- Silver, D.; Lever, G.; Heess, N.; Deen, T.; Wierstra, D.; Riedmiller, M.; and van Hasselt, H. 2014. Deterministic Policy Gradient Algorithms. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32, 387–395.
- Sima, C.; Renz, K.; Chitta, K.; Chen, L.; Zhang, H.; Xie, C.; Beisswenger, J.; Luo, P.; Geiger, A.; and Li, H. 2024. DriveLM: Driving with Graph Visual Question Answering. In *Computer Vision – ECCV 2024*, 256–274.
- Sutskever, I.; Vinyals, O.; and Le, Q. V. 2014. Sequence to Sequence Learning with Neural Networks. In *Advances in Neural Information Processing Systems*, volume 27.
- Sutton, R. S. 1988. Learning to Predict by the Methods of Temporal Differences. *Machine Learning*, 3: 9–44.
- Sutton, R. S.; and Barto, A. G. 1998. *Reinforcement Learning: An Introduction*. Cambridge, MA: MIT Press.
- Sutton, R. S.; Precup, D.; and Singh, S. 1999. Between MDPs and Semi-MDPs: A Framework for Temporal Abstraction in Reinforcement Learning. *Artificial Intelligence*, 112(1–2): 181–211.
- Thrun, S.; Montemerlo, M.; Dahlkamp, H.; Stavens, D.; Aron, A.; Diebel, J.; Fong, P.; Gale, J.; Halpenny, M.; Hoffmann, G.; et al. 2006. Stanley: The Robot That Won the DARPA Grand Challenge. *Journal of Field Robotics*, 23(9): 661–692.
- Treiber, M.; Hennecke, A.; and Helbing, D. 2000. Congested Traffic States in Empirical Observations and Microscopic Simulations. *Physical Review E*, 62(2): 1805–1824.
- Turri, V.; Besselink, B.; and Johansson, K. H. 2017. Co-operative Look-Ahead Control for Fuel-Efficient and Safe Heavy-Duty Vehicle Platooning. *IEEE Transactions on Control Systems Technology*, 25(1): 12–28.
- Turri, V.; Besselink, B.; Martensson, J.; and Johansson, K. H. 2014. Fuel-Efficient Heavy-Duty Vehicle Platooning by Look-Ahead Control. In *53rd IEEE Conference on Decision and Control*, 654–660.
- Turri, V.; Flardh, O.; Martensson, J.; and Johansson, K. H. 2018. Fuel-Optimal Look-Ahead Adaptive Cruise Control for Heavy-Duty Vehicles. In *2018 Annual American Control Conference (ACC)*, 1841–1848.
- van Hasselt, H.; Guez, A.; and Silver, D. 2016. Deep Reinforcement Learning with Double Q-learning. In *Proceedings of the 30th AAAI Conference on Artificial Intelligence*, 2094–2100.
- Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, volume 30.
- Watkins, C. J. C. H.; and Dayan, P. 1992. Q-Learning. *Machine Learning*, 8(3–4): 279–292.
- Werling, M.; Ziegler, J.; Kammel, S.; and Thrun, S. 2010. Optimal Trajectory Generation for Dynamic Street Scenarios in a Frenet Frame. In *Proceedings of the 2010 IEEE International Conference on Robotics and Automation*, 987–993.
- Williams, R. J.; and Zipser, D. 1989. A Learning Algorithm for Continually Running Fully Recurrent Neural Networks. *Neural Computation*, 1(2): 270–280.
- Wu, X.; Li, J.; Su, C.; Fan, J.; and Xu, M. 2023. A Deep Reinforcement Learning Based Hierarchical Eco-Driving Strategy for Connected and Automated HEVs. *IEEE Transactions on Vehicular Technology*, 1–16.
- Xu, Z.; Zhang, Y.; Xie, E.; Zhao, Z.; Guo, Y.; Wong, K.-Y. K.; Li, Z.; and Zhao, H. 2024. DriveGPT4: Interpretable End-to-End Autonomous Driving Via Large Language Model. *IEEE Robotics and Automation Letters*, 9: 8186–8193.
- Yurtsever, E.; Lambert, J.; Carballo, A.; and Takeda, K. 2020. A Survey of Autonomous Driving: Common Practices and Emerging Technologies. *IEEE Access*, 8: 58443–58469.
- Zeng, A.; Wong, A.; Welker, S.; Choromanski, K.; Tombari, F.; Purohit, A.; Sinha, M.; Lee, V.; Vanhoucke, V.; Xia, F.; et al. 2022. Socratic Models: Composing Zero-Shot Multimodal Reasoning with Language. *arXiv preprint arXiv:2204.00598*.